

Rockchip Rockit Runtime Library Developer Guide

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Rockchip Electronics Co., Ltd.

No.18 Building, A District, No.89, software Boulevard Fuzhou, Fujian, PRC

Website: www.rock-chips.com

Customer service Tel: +86-4007-700-590

Customer service Fax: +86-591-83951833

Customer service e-Mail: fae@rock-chips.com

Preface

Overview

This document presents a brief introduction to Rockit Runtime Library.

Product Version

Chipset	Kernel Version
RV1106/RV1103	Linux5.10
RV1103B/RV1106B	Linux5.10
RK3308	Linux6.1
RK3506	Linux6.1/rt-thread 4.1
RV1126B	Linux6.1

Intended Audience

This document (this guide) is mainly intended for:

Technical support engineers

Software development engineers

1. Revision History

Date	Version	Author	Change Description
2022-04-06	V1.4.11	SCH	Initial version
2022-05-26	V1.4.15	SCH	Add VENC and some description
2022-11-30	V1.4.30	SCH	Add log level setting
2023-07-27	V1.4.56	SCH	Added MCU KO loading parameter introduction; Added debugging parameter introduction
2024-10-16	V2.13.0	SCH	Update VSYS/VLOG/VALLOC, Add Cross-platform Instructions
2024-11-01	V2.14.2	CXX	Update VENC parameter introduction Add IVS parameter introduction Add mapping relationship between venc_info and rockit API
2025-4-27	V2.28.0	SCH	Add relevant descriptions of RK3308, RK3506 and RV1126B

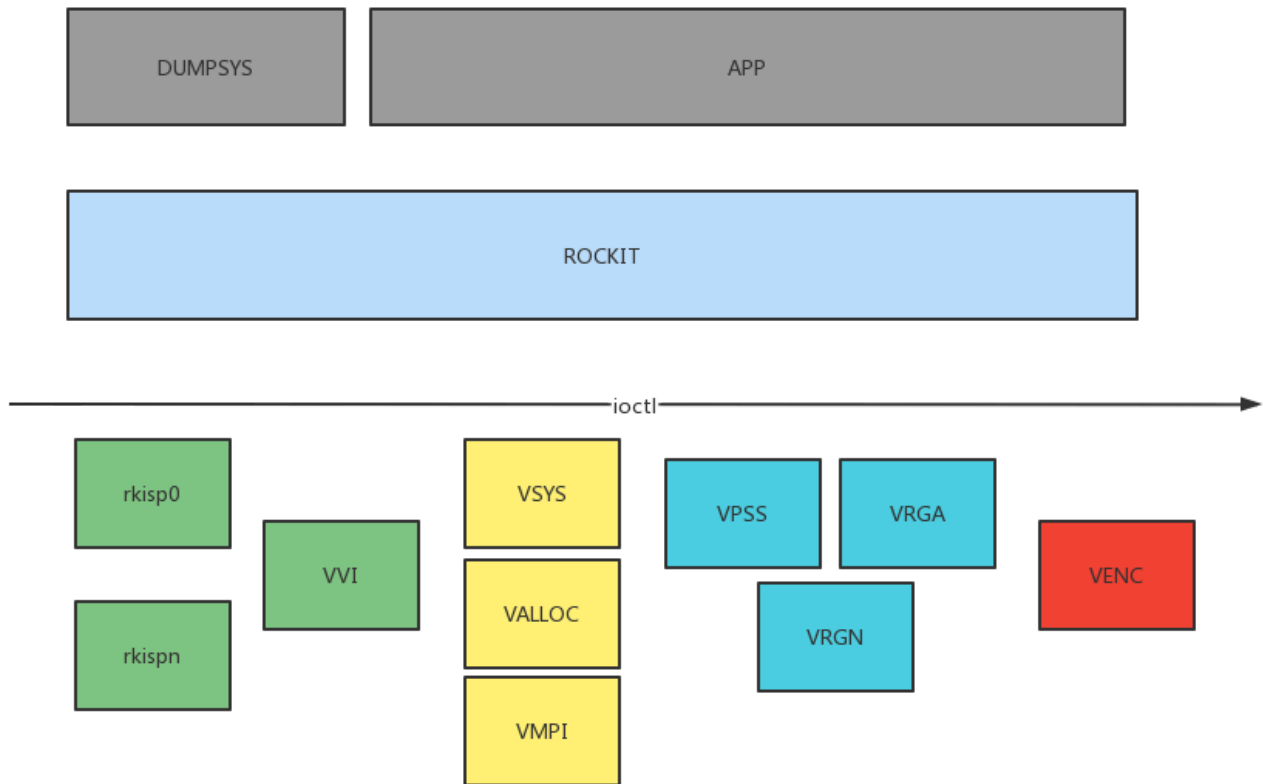
2. Contents

Rockchip Rockit Runtime Library Developer Guide

1. Revision History
2. **Contents**
3. Framework
 - 3.1 Parameters Supported by KO
 - 3.1.1 mcu_fw_path
 - 3.1.2 mcu_fw_addr
 - 3.1.3 isp_max_h
 - 3.1.4 isp_max_w
 - 3.1.5 en_mcu
 - 3.1.6 rk_dma_heap_base
 - 3.1.7 rk_dma_heap_size
4. Cross-platform Porting
5. Cross - platform Communication
6. VALLOC
7. VLOG
 - 7.1 Kernel LOG
 - 7.2 How to Set the LOG Output Level of the Module
 - 7.3 Set the Output Path and Size of SO LOG
 - 7.4 Clear LOG
8. VSYS
9. VENC
10. Mapping relationship between venc_info and rockit API
11. IVS
12. VRGN
13. VRGA

3. Framework

In order to facilitate Rockit porting, a new runtime libraries has been developed in the kernel. Hereinafter referred to as RRT. The block diagram of RRT in the kernel is as follows:



```
.
├─ isp.ko    ---- isp module, corresponding to the /dev/mpi/rkisp-vir0 node
├─ valloc.ko --- buffer manager, corresponding to the /dev/mpi/valloc node
├─ venc.ko    --- Encoder module, corresponding to the /dev/mpi/venc
├─ vmpi.ko    --- media framework, corresponding to the /dev/mpi/vmpi
├─ vpss.ko    --- vpss module, corresponding to the /dev/mpi/vpss
├─ vrga.ko    --- rga module, corresponding to the /dev/mpi/vrga
├─ vrgn.ko    --- rgn module, corresponding to the /dev/mpi/vrgn
├─ vsys.ko    --- system module, corresponding to the /dev/mpi/vsys
└─ vvi.ko     --- vi module, corresponding to the /dev/mpi/vvi
```

Before using rockit interface, the above ko need to be loaded, and the loading way is as follows (**note: all the above ko have been merged into one ko**):

```
insmod rockit.ko
```

Use the following command to check whether the loading is successful

```
# lsmod
Module                Size  Used by    Tainted: G
rknpu                 21535  1
rockit                90530  14
mpp_vcodec            350020  4 rockit
rga3                  76968  3 rockit
sc4336                8940  2
phy_rockchip_csi2_dphy 7657  0
phy_rockchip_csi2_dphy_hw 7644  0
video_rkisp           138585  3 rockit,[permanent]
video_rkcif           107889  4
rk_dvbm               5813  2 mpp_vcodec,video_rkisp
```

After loading successfully, you can run related programs that depend on rockit.so.

3.1 Parameters Supported by KO

3.1.1 mcu_fw_path

The path of MCU wrapping firmware, it is empty by default, does not load MCU firmware.

```
insmod rockit.ko mcu_fw_path=/oem/usr/ko/hpmcu_wrap.bin
```

3.1.2 mcu_fw_addr

The address of the MCU wrapping firmware, it is 0xff6ff000(rv1106)/0x210f6000(rv1103b) by default.

```
insmod rockit.ko mcu_fw_addr=0xff6ff000
```

3.1.3 isp_max_h

It is the height of the ISP front-end input. If VICAP has configured the cropping function, it should be configured the width after cropping. If the secondary parameter is configured incorrectly, cat /dev/mipi/vlog will prompt for the correct value.

```
mpi_log_err("isp input height config err, config h= %d, real h = %d", isp_max_h, isp_h);
```

```
insmod rockit.ko isp_max_h=1920
```

3.1.4 isp_max_w

Reserved

3.1.5 en_mcu

Whether to enable the MCU driver. It is enabled(rv1106)/disabled(rv1103b) by default. The MCU driver will not be loaded after it is disabled.

```
insmod rockit.ko en_mcu=0
```

3.1.6 rk_dma_heap_base

The first address of the MMZ released by RK_MPI_SYS_ReleasePhyMemory needs to be 4K aligned and larger than the starting address of MMZ. The calculation method of the MMZ starting address is: total physical memory - MMZ size.

```
insmod rockit.ko rk_dma_heap_base=0x40000
```

3.1.7 rk_dma_heap_size

The size of the MMZ released by RK_MPI_SYS_ReleasePhyMemory needs to be less than or equal to the size of the MMZ.

```
insmod rockit.ko rk_dma_heap_size=0x40000
```

4. Cross-platform Porting

The RRT has been enhanced with the addition of OSAL (Operating System Abstraction Layer) and rewritten based on OSAL, enabling it to run not only in the Linux kernel mode but also in the Linux user mode and on RTOS (Real-Time Operating Systems). Currently, there are two instances of the RRT in RV1103B/RV1126B, one running in user mode and the other in kernel mode. To view /dev/mpi/xxx related nodes on LINUX user mode and RTOS, the command dumpsys needs to be prefixed to the original command.

```
dumpsys cat /dev/mpi/valloc
```

```
dumpsys cat /dev/mpi/vsys
```

```
dumpsys cat /dev/mpi/vlog
```

```
dumpsys cat /dev/mpi/vrgn
```

5. Cross - platform Communication

Multiple RRTs may run on a single SOC. To reduce the communication cost for module developers, RRT has achieved cross - platform interoperability. Specifically, the VALLOC module is used to manage the reference count of cross - platform BUFFERS, and the VSYS module is used to implement cross - platform binding and the transfer of BUFFERS. Therefore, there is a prerequisite for cross - platform communication: the RRTs on each platform can share BUFFERS through physical memory.

The following table shows the RRTs supported by each SOC and the modules included in each RRT.

SOC	RRT (Linux Kernel Mode)	RRT (Linux User Mode)	RRT (rt - thread)
RV1106/RV1103	VI / VPSS / VENC / AVS / VRGN / VALLOC / VSYS	Not supported	Not supported
RV1103B/RV1106B	VI / VPSS / VENC / AVS / VRGN / VALLOC / VSYS	VDEC / VO / VRGN / AI / AO / AENC / ADEC / AMIX / AF / TDE / VALLOC / VSYS / TDE	Not supported
RK3308	Not supported	VALLOC / VSYS / VRGN / VDEC / AI / AO / ADEC / AENC / AMIX / AF / VO	Not supported
RK3506	Not supported	VALLOC / VSYS / VRGN / VDEC / AI / AO / ADEC / AENC / AMIX / AF / VO / TDE	VALLOC / VSYS / VRGN / VDEC / AI / AO / ADEC / AENC / AMIX / AF / VO / TDE
RV1126B	VI / VPSS / VENC / AVS / VRGN / VALLC / VSYS / GDC / IVS	VI / VALLOC / VSYS / VRGN / VDEC / AI / AO / ADEC / AENC / AMIX / AF / VO / TDE	Not supported

6. VALLOC

The memory allocator of the media framework, which is responsible for allocating and managing the memory required by the media framework. You can use the following command to check the current memory status:

```
cat /dev/mpi/valloc
```

Detailed Information:

Item	Description	Remarks
module	Module information	Module name
version	Version information	Repository name: commit: version number
build	Build information	Compiler and Compilation date
mpi buf list	Buffer list separator	The following information is related to BUF
buf_id	Unique ID of this platform	Used to identify MPI BUF, equivalent to HANDLE. It will display two IDs, with the ID in parentheses being the cross-platform unique ID
pool	Buffer pool ID	The buf pool to which the mpi buf belongs, -1 means no pool
ref	MPI buf usage count	+1 before use, -1 after use, 0 means no pool, mpi buf will be released
dma	DMA buf usage count	+1 before use, -1 after use, 0 means dma_buf will be released
use	Reference count added by non - this RRT	When ref - use == 0, it indicates that this RRT does not use this BUF.
basefd	Basic fd of the dma buf	For RRT in the Linux kernel mode, basefd = -1
status	Current status of the buf	bit0: 1 - indicates that the object has been registered, 0 - the resource has been released bit1: 1 - indicates that the object is accessible, 0 - inaccessible bit2: 1 - indicates that the object has been deleted, 0 - not deleted bit3: 1 - indicates that the object is not allowed to be deleted, 0 - allowed to be deleted bit4: 1 - indicates a private object, 0 - a public object bit5: 1 - indicates that the ID is retained when the object is deleted, 0 - not retained bit6: 1 - indicates that the object can be monitored, 0 - cannot be monitored bit7-9: Reserved bit10: 1 - indicates that the BUFFER has been deleted bit11-31: Reserved
size	Buffer size	Unit: BYTE, 0 means it is in wrapping mode
handle	Buffer handle	For RV1106, it is the physical address
vaddr	Vitrual Adress	RTOS Virtual Address (Same as Physical Address)
paddr	Physical Adress	Peripheral address on the IC bus
create	Buffer creator	Records the distribution of BUFFER, "deleted" means that the creator of this BUFFER has been deleted.
user	Buffer user	For all rotating BUFFERS, the current user will be recorded. and there will be POOLS in the rotating BUFFERS
total	Total memory	Unit: BYTE
mpi pool list	Buffer pool list separator	The following information is related to POOL
poid	Global unique ID	pool_id = poid means that this BUFFER belongs to this POOL
total	Total number of BUFFERS	

Item	Description	Remarks
free	Remaining number of BUFFERS	Currently available BUFFERS
min	Minimum remaining	-1 indicates frame loss
size	BUFFER SIZE in the pool	All BUFFER sizes must be larger than this value
reflevel	POOL Idle BUF REFCNT Watermark	MPI BUF is considered idle when its REFCNT is less than or equal to this value
status	Current pool status	bit0: 1-Indicates the object is registered, 0-Resource is released bit1: 1-Indicates the object is accessible, 0-Inaccessible bit2: 1-Indicates the object is deleted, 0-Not deleted bit3: 1-Indicates the object is not allowed to be deleted, 0-Deletion allowed bit4: 1-Indicates a private object, 0-Public object bit5: 1-Indicates the ID is retained upon deletion, 0-Not retained bit6: 1-Indicates support for monitoring, 0-Not monitorable bit7-31: Reserved
create	Pool creator	
user	Pool user	

7. VLOG

LOG manager of media framework.

7.1 Kernel LOG

```
cat /dev/mpi/vlog
```

7.2 How to Set the LOG Output Level of the Module

The command `echo module=level > /tmp/rt_log_level` is used to change the LOG output level of the module, and this command affects rockit so and ko at the same time.

module = all cmpi mb sys vdec venc rgn vpss vgs tde avs wbc vo vi ai ao aenc adec

level=0,1,2,3,4,5,6; 2 is ERR (red), 3 WRN (yellow), 4 INF (green).

7.3 Set the Output Path and Size of SO LOG

```
echo rt_log_path=/tmp/rockit.log > /tmp/rt_log_level
```

```
echo rt_log_size=1024000 > /tmp/rt_log_level
```

7.4 Clear LOG

```
echo 0 > /dev/mpi/vlog
```

8. VSYS

The manager for media relationship binding, you can check the current binding relationship and frame rate information through the following command:

```
cat /dev/mpi/vsys
```

Detailed Information:

Project	Description	Remarks
module	module information	module name
version	version information	warehouse name: commit: version number
build	Compilation information	Compiler-Compilation date
mpi node list	node list dividing line	The following is node related information
id	Globally unique ID	Used to identify MPI node, equivalent to HANDLE
name	The name of the device used to create the NODE	Multiple NODEs may have the same name
handle	NODE Handle	Memory pointer to the node structure
nid	rockit custom ID	Custom ID to facilitate ROCKIT management NODE
uid	Data Flow Tree Node ID	This ID is used to query the MPI BUF user array, enabling the determination of the current owner of the BUF.
ref	Usage count	+1 before use, -1 after use, 0, release
infa_cnt	Count of Received frame	Total number of received frames of node
lcnt	Count of Received frame loss	Frame loss caused by inability to process or processing failure, frame rate control is not within this range
frate	Received frame rate	Statistics in S, setting is not supported
fbase	Received frame loss ratio	Receive fbase frame, take the first frame, and lose the rest
wcnt	The actual frame rate of the user queue on the receiving end	User sets the user queue size through depth, and the frames that are not released after the user obtains them are also counted
depth	The depth of user queue on the receiving end	Cannot exceed the total size of the front-end BUF queue, otherwise the front-end may be stuck
timeout	Timeout period on the receiving end to obtain user queue data	Unit: ms
onfa_cnt	Count of sent frame	Total number of sent frames of node
lcnt	Count of sent frame loss	Frame loss caused by inability to process or processing failure, frame rate control is not within this range
frate	Sent frame rate	Statistics in s, setting is not supported
fbase	Sent frame loss ratio	Send fbase frames, send the first frame, and lose the rest
wcnt	The actual frame rate of the user queue on the sending end	User sets the user queue size through depth, and the frames that are not released after the user obtains them are also included
depth	The depth of user queue on the sending end	Cannot exceed the total size of the front-end BUF queue, otherwise the front-end may be stuck
timeout	Timeout period of the sending end to obtain user queue data	Unit: ms
itime	Time taken to receive 1 frame	Unit: us

Project	Description	Remarks
otime	Time taken to send 1 frame	Unit: us
state	Current node status	bit0: 1-Indicates the object is registered, 0-Resource is released bit1: 1-Indicates the object is accessible, 0-Inaccessible bit2: 1-Indicates the object is deleted, 0-Not deleted bit3: 1-Indicates the object is not allowed to be deleted, 0-Deletion allowed bit4: 1-Indicates a private object, 0-Public object bit5: 1-Indicates the ID is retained upon deletion, 0-Not retained bit6: 1-Indicates support for monitoring, 0-Not monitorable bit7-9: Reserved bit10: 1-In deletion process... bit11: 1-Refreshing BUF... bit12: 1-In operation... bit13: 1-Has a binding with a preceding stage... bit14: 1-Automatically operates upon binding with a preceding stage... bit15-31: Reserved
next node	Next NODE ID	Supports 1 NODE binding multiple NODEs
mpi node frame control	Frame rate control table of the node	The following is information related to NODE frame rate control
id	Globally unique ID	Used to identify MPI node, equivalent to HANDLE
name	The name of the device used to create the NODE	Multiple NODEs may have the same name
table	Fame loss table	The size is the frame rate, 0 means to keep, 1 means to lose
mpi dev list	ROCKIT Device List	Below are the device information
id	Device ID	Globally unique identifier
name	Device Name	Used for easy differentiation of devices
handle	Device Handle	Memory address of the device structure
ref	Reference Count	Incremented by 1 for each open handle reference
node	NODE on Device	Lists all NODEs on the device by their IDs

9. VENC

Encoding parameters, can be obtained by the following command:

```
cat /proc/vcodec/enc/venc_info
```

- venc thread status

Parameter	Description
last_running	The time when the thread most recent wake-up
run_cnt	Number of times the thread has run
que_cnt	Number of times the thread has been triggered

- venc chn running status

Parameter	Description
ID	Encoding channel ID
running	Whether the channel is active
combo_run	Whether JPEG combined encoding in a channel is active
cfg_gap	Time consumption of registers of a channel configuration(in ms)
strm_cnt	Number of cached streams in a channel
strm_out	Number of streams sent to the application in a channel
gap_time	Interval between the last two encoding start times of a channel (in ms)
cb_gap_time	Time interval between two JPEG combined encoding operations
last_cb_start	The most recent start time of the JPEG combined encoding.
last_cb_end	The most recent end time of the JPEG combined encoding.
queue_size	Buffer queue size in the channel
pkt_total_cnt	The total number of packets in a channel
pkt_user_get	Number of packets acquired by the application in the channel
pkt_user_put	Number of packets released by the application in the channel
seq_user_get	Seq of the current stream acquired by the application
seq_user_put	Seq of the current stream released by the application
seq_encoded	Seq of the current encoded content in the channel
seq_encoding	Seq of the content currently being encoded in the channel

- venc chn attr 1

Parameters	Description
ID	Encoding channel ID
Width	The width of encoding channel
Height	The height of encoding channel
Type	The type of encoding channel type
ByFrame	Get stream ID by frame n: get by package(currently not supported) y: get by frame
Sequence	Sequence ID It is the frame sequence ID when it is acquired by frame, and it is the package sequence ID when it is acquired by package
GOPMode	GOP mode
Prio	The value of the channel priority
MaxWidth	The max width of encoding channel
MaxHeight	The max height of encoding channel
Online	Warp enable switch
RefShare	Whether the reconstruction/reference is shared

- venc chn attr 2

Parameters	Description
ID	Encoding channel ID
Vestr	Whether to enable encoding
SrcFr	The source frame rate of VENC frame rate control, that is, the input frame rate
TarFr	Target frame rate of VENC frame rate control
Timeref	VENC start time
PixFmt	The format of encoding the current frame
RealFps	VENC real frame rate
Rotation	Whether VENC enables rotation
Mirror	Whether VENC enables mirroring

- ring buf status

Parameters	Description
ID	Encoding channel ID
w_pos	Write position of code stream buf
r_pos	Read position of code stream buf
used_len	Space used by stream buf
total_len	The total space size of code stream buf
min_size	Latest free space required for single frame allocation
l_w_pos	Last updated write position
l_r_pos	Last updated read position
max_use_len	The maximum space used by code stream buf

- hw status

Parameters	Description
ID	Encoding channel ID
hw_run	The current channel is running
enc_status	enc status: 0: ENC_STATUS_CFG_IN, 1: ENC_STATUS_CFG_DONE, 2: ENC_STATUS_START_IN, 3: ENC_STATUS_START_DONE, 4: ENC_STATUS_RUN_TASK_IN, 5: ENC_STATUS_RUN_TASK_DONE, 6: ENC_STATUS_INT_IN, 7: ENC_STATUS_INT_DONE, 8: ENC_STATUS_SLICE_INFO_IN, 9: ENC_STATUS_SLICE_INFO_DONE
pkt_fail_cnt	Times of enc request packet pool failures
ring_fail_cnt	Times of stream buf allocation failures
cfg_fail_cnt	Times of generating hardware configuration information failures
start_fail_cnt	Times of enabling hardware encoding failures
pkt_full_cnt	The number of frame drops caused by PACKET exceeding 100.

- hw param

Parameters	Description
ID	Encoding channel ID
qp_dlt_row_i	The maximum amplitude value of row-level QP adjustment for I-frames
qp_dlt_row	The maximum amplitude value of row-level QP adjustment for P-frames
qbias_en	Bias adjustment switch
qbias_i	Bias value of I block
qbias_p	Bias value of P block
flt_str_i	Filter strength of I-frames
flt_str_p	Filter strength of P-frames
thrd_i	Block-level madi threshold table for I-frames
thrd_p	Block-level madi threshold table for P-frames
step_i	Table used for AQ lookup based on thrd_i for I-frames
step_p	Table used for AQ lookup based on thrd_p for P-frames
qbias_arr	Bias configuration
aq_rnge_arr	Configuration of AQ constraint range

- fine tuning param

Parameters	Description
ID	Encoding channel ID
scene_mode	Scene mode
md_swth_en	Motion static switch
deblur_str	Strength of deblur
atr_str	Strength of anti-ring
atl_str	Strength of anti-line
lambda_idx	Lambda index for P-frames
lambda_i_idx	Lambda index for I-frames
atf_str	Strength of anti-flicker
static_frm_num	Threshold value of frames to determine block stillness
madp16_th	Threshold of madp16 for block stillness
skip16_wgt	Enhancement effect of skip16 bias
skip32_wgt	Enhancement effect of skip32 bias
lgt_chg_lvl	Optimization parameter level for scenes with lighting changes

- venc roi attr

Parameters	Description
ID	Encoding channel ID
roi	Index of ROI
x	Starting coordinate in the horizontal direction
y	Starting coordinate in the vertical direction
w	Width of ROI
h	Height of ROI
quality	QP value of ROI
intra	Whether ROI is forcibly set to intra
abs_qp	Absolute QP enable switch

- hw session infos

Parameters	Description
ID	Encoding channel ID
device	Device name
hw_running	Hardware running status
online	Warp enable switch
wrap_ovfl	Warp overflow count
bs_ovfl	Stream buffer overflow count
wrap_src_mis	Warp source mismatch count
enc_err_cnt	Encoder error count
enc_err_irq	Interrupt register value when the encoder error
enc_err_st	Status register value when the encoder error

- bitstream buffer addr

Parameters	Description
top	Top of the stream buffer stack
bot	Bottom of the stream buffer stack
wr	Write address of the stream buffer
rd	Read address of the stream buffer

- h264e chn attr

Parameters	Description
ID	Encoding channel ID
Width	Width in pixels
Height	Height in pixels
Profile	The profile of encoding channel Base: baseline Mp:main profile Hp:high profile

- Syntax INFO1

Parameters	Description
ID	Encoding channel ID
Slcspalten	Whether to use slice split
SplitMode	slice split mode 1: split by byte 2: split by macroblock
Slcsize	The size of the slice, when the split_mode is different, the meanings of slice_size is different
IntraRefresh	Whether to enable P frame to refresh I macroblocks
RefreshMode	The mode of refreshing I macroblocks row: Refresh by row column: Refresh by column
RefreshNum	Refresh the number of rows or columns of I macroblocks in each frame

- Syntax INFO2

Parameters	Description
ID	Encoding channel ID
Profile	The profile type of encoding channel Base: Baseline MP: Main profile HP: High profile
Cabac	Entropy encoding mode, cavlc or cabac
Trans8	Transformation mode used Only trans4x4 mode is enabled by default Y: enable trans8x8 mode
ScalingMode	Scaling list mode
Poc	The value of the syntax element pic_order_cnt_type
Alpha	The value of the syntax element slice_alpha_c0_offset_div2
Beta	The value of the syntax element slice_beta_offset_div2

- h265e chn attr

Parameters	Description
ID	Encoding channel ID
Width	Width in pixels
Height	Height in pixels
Profile	The profile of encoding channel main: main profile

- Syntax INFO1

Parameters	Description
ID	Encoding channel ID
Slcsplten	Whether to use slice split
SplitMode	slice split mode 1: split by byte 2: split by macroblock
Slcsize	The size of the slice, when the split_mode is different, the meanings of slice_size is different
IntraRefresh	Whether to enable P frame to refresh I macroblocks
RefreshMode	The mode of refreshing I macroblocks row: refresh by row column: refresh by column
RefreshNum	Refresh the number of rows or columns of I macroblocks in each frame

- Syntax INFO2

Parameters	Description
ID	Encoding channel ID
Dblken	Negated value of syntax element slice_deblocking_filter_disabled_flag
Tc	The value of syntax element slice_tc_offset_div2
Beta	The value of the syntax element slice_beta_offset_div2
Saoluma	The value of the syntax element slice_sao_luma_flag
Saochroma	The value of the syntax element slice_sao_chroma_flag
IntraSmoothing	The value of the syntax element strong_intra_smoothing_enabled_flag
ScalingMode	Scaling list mode
TmvpEn	TMVP enable switch

- Trans Info

Parameters	Description
ID	Encoding channel ID
CbQpoffset	The value of syntax element pps_cb_qp_offset
CrQpoffset	The value of syntax element pps_cr_qp_offset

- jpeg chn attr

Parameters	Description
ID	Encoding channel ID
width	Image width in Pixel
height	Image height in Pixel
quant	Channel quantization qp
qfactor	The qfactor used by channel
qfmax	The maximum value of qfactor used by channel
qfmin	The minimum value of qfactor used by channel

- RC base param1

Parameter	Description
ChnId	Encoding channel ID
Gop	Encoding Group of Pictures (GOP)
StatTm	Rate control statistical time in seconds
ViFr	VI (Video Input) frame rate
TrgFr	Encoding target frame rate
RcMode	Rate control mode (cbr/fixqp/vbr/avbr/smart)
Br(kbps)	Bitrate in kbps
IQp	QP of I-frames, valid in FixQp mode
PQp	QP of P-frames, valid in FixQp mode

- RC base param2

Parameter	Description
ChnId	Encoding channel ID
MinQp	Minimum QP for P-frames
MaxQp	Maximum QP for P-frames
MinIQp	Minimum QP for I-frames
MaxIQp	Maximum QP for I-frames
EnableIdr	IDR enable switch Y: Enabled N: Disabled

- RC run comm param1

Parameter	Description
ChnId	Encoding channel ID
bLost	Whether to drop frames when the instantaneous bitrate exceeds the set threshold
LostThr	The threshold of frame dropping
LostFrmstr	Statistics of the number of frame dropping
EncGap	Gap between encoding frames

- RC run comm param2

Parameter	Description
ChnId	Encoding channel ID
SprFrmMod	Super-sized frame processing mode
SprIFrm	Threshold of super-sized I-frames
SprPFrm	Threshold of super-sized P-frames
RCPriority	Rate control priority bitrate: Target bitrate priority framebits: Super-sized frame threshold priority

- RC gop mode attr

Parameter	Description
ChnId	Encoding channel ID
GopMode	Type of gop_mode
IpQpDelta	The Qp Delta value of IP frame. For SmartP mode, it displays the Qp Delta value of Bg frames and P-frames. Range: [-10, 30]
BgInterval	Interval of special P-frames Range: Less than or equal to Gop
ViQpDelta	The Qp Delta value of virtual I-frames and P-frames Range: [-10, 30]

- RC run vbr common param

Parameter	Description
ChnId	Encoding channel ID
ChgPs	The ratio of the bitrate at the start of VBR QP adjustment to the maximum bitrate. This value is calculated by $\text{bitrate}/\text{max_bitrate} * 100$
MinIprop	The minimum ratio of IP frame bitrate. Range: [1,100]. Default value: 1
MaxIprop	The maximum ratio of IP frame bitrate. Range: [MinIprop, 100]. Default value: 100
MaxQp	The maximum QP value of P frames. Range: [0, 51]. Default value: 51
MinQp	The minimum QP value of P frames. Range: [0, MaxQp]. Default value: 24
MaxIQp	The maximum QP value of I frames. Range: [0, 51]. Default value: 51
MinIQp	The minimum QP value of I frames. Range: [0, MaxIQp]. Default value: 24
FrmMaxQp	Maximum QP for P-frames at the frame level
FrmMinQp	Minimum QP for P-frames at the frame level
FrmMaxIQp	Maximum QP for I-frames at the frame level
FrmMinIQp	Minimum QP for I-frames at the frame level
MaxReEncTimes	Number of re-encoding times per frame. 0 means no re-encoding. Range: [0, 3] Default value: 1

- RC run cbr param

Parameter	Description
ChnId	Encoding channel ID
MinIprop	The minimum ratio of IP frames. Range: [1, 100]
MaxIprop	The maximum ratio of IP frames. Range: [MinIprop, 100]
MaxQp	The maximum QP of P-frames, used to clamp quality. Range: [0, 51]
MinQp	The minimum QP of P-frames, used to clamp the fluctuation of bitrate. Range: [0, MaxQp]
MaxIQp	The maximum QP of I-frames, used to control the minimum number of bits for I-frames. Range: [0, 51]
MinIQp	The minimum QP of I-frames, used to control the maximum number of bits for I-frames. Range: [0, MaxIQp]
FrmMaxQp	Maximum QP for P-frames at the frame level
FrmMinQp	Minimum QP for P-frames at the frame level
FrmMaxIQp	Maximum QP for I-frames at the frame level
FrmMinIQp	Minimum QP for I-frames at the frame level
MaxReEncTimes	Number of re-encoding times per frame. 0 means no re-encoding. Range: [0, 3]

- RC run avbr param

Parameter	Description
ChnId	Encoding channel ID
MaxStillQP	Maximum still QP
MotionSensi	Motion sensitivity
MinPercent	Minimum still percentage

- RC HierarchicalQp INFO

Parameter	Description
ChnId	Encoding channel ID
bEnable	Whether layered QP is enabled
FrameNum[0]	Number of frames of QP in layer 1
FrameNum[1]	Number of frames of QP in layer 2
FrameNum[2]	Number of frames of QP in layer 3
FrameNum[3]	Number of frames of QP in layer 4
QpDelta[0]	QpDelta of frames in layer 1
QpDelta[1]	QpDelta of frames in layer 2
QpDelta[2]	QpDelta of frames in layer 3
QpDelta[3]	QpDelta of frames in layer 4

- RC debreath_effect info

Parameter	Description
ChnId	Encoding channel ID
bEnable	Whether to enable breath removal
Strength0	Breath removal strength 0
DeBrthEfctCnt	Times of breath removal which is applied to the image

- RC run CBR/VBR info1

Parameter	Description
ChnId	Encoding channel ID
InsBr(kbps)	Instantaneous bitrate in kbps
InsFr	Instantaneous frame rate
WatL	Bitrate threshold
CfgBt(kb)	Target size of the previous frame in kb
RealBt(kb)	Actual size of the previous frame in kb
IPRatio	The ratio between Average size of I-frames and P-frames
StartQp	Starting QP
MinQp	Minimum QP
MaxQp	Maximum QP

- mpp_buffer pool

Parameter	Description
unused_cnt	Number of unused packets
used_cnt	Number of packets in use
total_cnt	Total number of packets
max_cnt	Maximum number of packets

- mpp_buffer pool

Parameter	Description
unused_cnt	Number of unused buffer
used_cnt	Number of buffer in use
total_cnt	Total number of buffer
max_cnt	Maximum number of buffer

- mpp_frame pool

Parameter	Description
unused_cnt	Number of unused frames
used_cnt	Number of frames in use
total_cnt	Total number of frames
max_cnt	Maximum number of frames

10. Mapping relationship between venc_info and rockit API

venc_info is the encoder configuration parameters printed by the kernel. currently only RV1103B/RV1106B and RV1103/RV1106 are supported. This chapter primarily focuses on key configurations, emphasizing subjective effects and objective compression-related settings.

venc_info	rockit api	rockit api member	note
qbias_en qbias_i qbias_p	RK_MPI_VENC_SetH264Qbias RK_MPI_VENC_SetH265Qbias	bEnable u32QbiasI u32QbiasP	RV1103/RV1106 supported, RV1103B/RV1106B not supported
qbias_en qbias_arr[18]	RK_MPI_VENC_SetH264Qbias2 RK_MPI_VENC_SetH265Qbias2	bEnable u32Qbias2[18]	RV1103/RV1106 not supported, RV1103B/RV1106B upgrade version
StartQp MaxQp MinQp MaxIQp MinIQp IpQpDelta MaxReEncTimes FrmMaxQp FrmMinQp FrmMaxIQp FrmMinIQp md_switch_qp	RK_MPI_VENC_SetRcParam	s32FirstFrameStartQp u32MaxQp u32MinQp u32MaxIQp u32MinIQp s32DeltIpQp s32MaxReEncodeTimes u32FrmMaxQp u32FrmMinQp u32FrmMaxIQp u32FrmMinIQp u32MotionStaticSwitchFrmQp	Inial QP,Block-level QP,Line-level QP configuration,Frame-level QP, Number of re-encode
thrd_i[16] thrd_p[16] step_i[16] step_p[16] qp_dlt_row_i qp_dlt_row	RK_MPI_VENC_SetRcParam2	u32ThrdI[16] u32ThrdP[16] s32AqStepI[16] s32AqStepP[16] u32RowQpDelta u32RowIQpDelta	AQ parameter, Line-level QP adjustment
aq_rnge_arr[10]	RK_MPI_VENC_SetRcParam3	u32AqRange[10]	RV1103/RV1106 not supported, RV1103B/RV1106B upgrade version
scene_mode	RK_MPI_VENC_SetSceneMode	enSceneMode	Scene mode
flt_str_i flt_str_p	RK_MPI_VENC_SetFilter	u32StrengthI u32StrengthP	Filter strength
md_swth_en	RK_MPI_VENC_EnableMotionStaticSwitch	bEnable	Motion static switch
deblur_str	RK_MPI_VENC_SetMotionDeblurStrength	u32Strength	Strength of deblur
atr_str	RK_MPI_VENC_SetAntiRing	u32AntiRing	Strength of anti-ring
atl_str	RK_MPI_VENC_SetAntiLine	u32AntiLine	Strength of anti-line
atf_str	RK_MPI_VENC_SetAntiFlick	u32AntiFlick	Strength of anti-flicker
lambda_i_idx lambda_idx	RK_MPI_VENC_SetLambda	u32LambdaI u32Lambda	Lambda configuration
static_frm_num madp16_th skip16_wgt skip32_wgt	RK_MPI_VENC_SetStaticWeight	u32FrmNum u32Madp16Th u32Skip16Weight u32Skip32Weight	Static scene skip tendency

lgt_chg_lvl	RK_MPI_VENC_SetLightChange	u32Level	RV1103/RV1106 not supported, RV1103B/RV1106B supported
DbkEn alpha beta	RK_MPI_VENC_SetH264Dbk	disable_deblocking_filter_idc slice_alpha_c0_offset_div2 slice_beta_offset_div2	H.264 deblocking filter
DbkEn Tc Beta	RK_MPI_VENC_SetH265Dbk	slice_deblocking_filter_disabled_flag slice_tc_offset_div2 slice_beta_offset_div2	H.265 deblocking filter
Saoluma Saochroma	RK_MPI_VENC_SetH265Sao	slice_sao_luma_flag slice_sao_chroma_flag	H.265 SAO parameters
IntraSmoothing	RK_MPI_VENC_SetH265PredUnit	strong_intra_smoothing_enabled_flag	H.265 intra-frame strong filtering
TmvpEn	RK_MPI_VENC_EnableTmvp	bEnable	H.265 TMVP switch
StatTm GopMode Gop	RK_MPI_VENC_SetChnAttr	u32StatTime enGopMode u32Gop	Rate control statistical time, GOP mode, GOP length
ScalingMode CbQpOffset CrQpOffset	RK_MPI_VENC_SetH265Trans	bScalingListEnabled cb_qp_offset cr_qp_offset	H.265 quantization parameters
trans8 ScalingMode CbQpOffset CrQpOffset	RK_MPI_VENC_SetH264Trans	u32TransMode bScalingListValid chroma_qp_index_offset	H.264's CbQpOffset and CrQpOffset reuse chroma_qp_index_offset
RcMode MinBr(kbps) MaxBr(kbps)	RK_MPI_VENC_SetChnAttr RK_MPI_VENC_EnableSvc	enRcMode u32MinBitRate u32MaxBitRate	Set enRcMode to VBR, configure u32MinBitRate and u32MaxBitRate, enable SVC to trigger smart mode
bEnable Strength0	RK_MPI_VENC_SetDeBreathEffect	bEnable s32Strength0	debreath parameters
cu_dqp	RK_MPI_VENC_SetH265CuDqp	u32CuDqp	H.265 diff_cu_qp_delta_depth

11. IVS

Intelligent video surveillance parameters can be accessed via the following command:

```
cat /proc/vcodec/ivs/ivs_info
```

- ivs channel attr

Parameter	Description
ID	Intelligent video surveillance channel ID
width	Channel width
height	Channel height
smear_en	smear enable switch
weightp_en	weightp enable switch
max_width	Maximum channel width
max_height	Maximum channel height
md_en	Motion detection enable switch
od_en	Occlusion detection enable switch

- ivs common cfg

Parameter	Description
ID	Intelligent video surveillance channel ID
fmt	Image format type
frm_acc_intval	Frame accumulation interval
frm_acc_gop	Frame accumulation GOP

- ivs md cfg

Parameter	Description
ID	Intelligent video surveillance channel ID
switch_sad	Motion detection SAD calculation mode selection 0: SAD 1: compensate SAD 2: the smaller of SAD and compensate SAD 3: the bigger of SAD and compensate SAD
thres_sad	4x4 block-level motion SAD threshold
thres_move	8x8 block motion judge threshold
night_mode	Night mode enable switch
filter_sw	Flying catkin filter enable switch
thres_dust_mv	The number threshold of flying catkins blocks around the current 8x8 block of the eight surrounding 8x8 blocks
thres_dust_blk	The number threshold of flying catkins blocks within the current 8x8 block consisting of four 4x4 blocks
thres_dust_chng	Threshold for the luminance mean difference between the current 8x8 block and the corresponding block at the previous frame position

- ivs od cfg

Parameter	Description
ID	Intelligent video surveillance channel ID
thres_complex	Complex threshold in occlusion detection
thres_complex_cnt	Complex threshold count in occlusion detection
thres_sad	SAD threshold

12. VRGN

The RGN manager of the media framework. Use the following command to see the current RGN information: `cat /dev/mpi/vrgn`

Detailed Information:

Item	Description	Remarks
module	Module information	Module name
version	Version information	Repository name: commit: version number
build	Build information	Compiler - compilation date
rgn list	RGN list separator	The following is the RGN-related information
handle	RGN handle	Used to identify RGN
type	RGN type	Used to identify RGN type: OVERLAY, COVER, MOSAIC
width	RGN area width	
height	RGN area height	
format	RGN area format	OVERLAY and COVER types are valid, "-1" indicates that the input type is not supported
buf_id	Currently used buffer ID for RGN	Valid for OVERLAY type, "-1" for other types
refcnt	MPI buffer usage count	
channel_num	Number of RGN attaches	Number of RGN ATTACH channels (modules)
node_rgn list	node_rgn list delimiter	The following is the NODE_RGN-related information
id	Unique ID for the node	Used to identify MPI node
name	Device name for creating the node	Multiple nodes may be with the same name
h	RGN handle	Used to identify the RGN, similar to HANDLE
x	RGN area x coordinate	X offset of the starting position of RGN
y	RGN area y coordinate	Y offset of the starting position of RGN
h x y	RGN area position information	Group of h, x, and y values that identify the RGN position. It is 8 groups in total.

13. VRGA

The RGA driver manager in Rockit. You can use the following command to check the current status of the RGA NODE:

```
cat /dev/mpi/vrga
```

The information description is as follows:

Item	Description	Remarks
id	Globally unique ID	Used to identify the MPI node, equivalent to the HANDLE
name	Device name for creating the NODE	Multiple NODEs may have the same name
ctx	Control handle of the rga node	Virtual address. It can be compared with the LOG to confirm which NODE has a problem
nid	Custom ID in Rockit	Custom ID, which facilitates Rockit to manage the NODE
ref	Usage count	Add 1 before use, subtract 1 after use. When it is 0, release it
bypass	Bypass flag	1 indicates that the RGA node does not perform any processing and directly passes through the data
csnt	Count of CROP parameter settings	The position of entering the queue of CROP parameter queue
cgnt	Count of obtaining CROP parameters	The position of exiting the queue of CROP parameter queue
ssnt	Count of SCALL parameter settings	The position of entering the queue of SCALL parameter queue
sgnt	Count of obtaining SCALL parameters	The position of exiting the queue of SCALL parameter queue
type	Type of RGA task	0: General, 1: OSD
swidth	Output width	The actual width of the output image
sheight	Output height	The actual height of the output image
svwidth	Output memory width	The memory width occupied by the output image, used for memory alignment.
svheight	Output memory height	The memory height occupied by the output image, used for memory alignment.
sfmt	Output image format	See in detail: rk_comm_video.h
cenalbe	Cropping enable flag	1: Indicates that cropping is performed on the input source
coordinate	Cropping parameters as a percentage	1: Indicates a percentage. 0: Absolute coordinates.
cxpos	X coordinate of the cropping area	When coordinate = 1: Indicates a percentage. When coordinate = 0: Absolute coordinates.
cypos	Y coordinate of the cropping area	When coordinate = 1: Indicates a percentage. When coordinate = 0: Absolute coordinates.
cwidth	Width of the cropping area	When coordinate = 1: Indicates a percentage. When coordinate = 0: Absolute width.
cheight	Height of the cropping area	When coordinate = 1: Indicates a percentage. When coordinate = 0: Absolute height.
job	Jobs under the current NODE	Lists the unprocessed jobs under the current NODE in the form of IDs